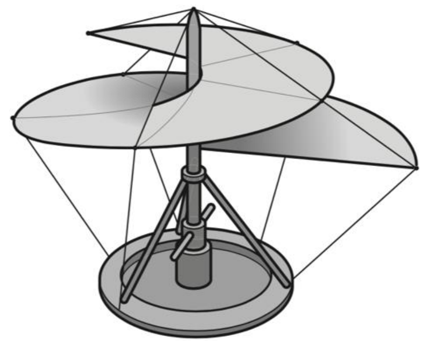
8. Leonardo da Vinci proposed a flying machine that would work like a screw to lift the pilot into the air. The 'screw' is rotated by the pilot.



The machine and the pilot together have a total mass of 120 kg.

Which useful output power must the pilot provide to move vertically upwards at a constant speed of 2.5 m s^{-1}?

A 48 W **B** 300 W **C** 470 W **D** 2900 W

Answer: D

[Method 1]

upward force provided by pilot to move at constant speed = weight

$$\text{power} = \text{force} \times \text{velocity} = 120 \times 9.81 \times 2.5 = 2943 \text{ W}$$

[Method 2]

$$\text{gain in g.p.e.} = mgv \times \text{time}$$

$$\text{power supplied} = \frac{\text{gain in g.p.e.}}{\text{time}} = mgv = 120 \times 9.81 \times 2.5 = 2943 \text{ W}$$